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Haoyang Yu (于浩洋)

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Education

Tsinghua University, Department of Statistics and Data Science Sept 2022 – June 2027

Ph.D. in Statistics Advisor: Hanzhong Liu (Expected)

Tsinghua University, Department of Physics Sept 2018 – June 2022

B.S. in Mathematics and Physics, Minor in Statistics

Research Interests

Causal inference; design-based theory; experimental design; network interference; high-dimensional covariate adjustment.

Publications

$*$ indicates corresponding authorship.

1. **Haoyang Yu**, Ke Zhu* and Hanzhong Liu (2025). Sharp Variance Estimator and Causal Bootstrap in Stratified Randomized Experiments. *Statistics in Medicine*, 44(13–14): e70139. [Paper] [Package]
2. Xin Lu, Wanjia Fu, Hongzi Li, **Haoyang Yu**, Honghao Zhang, Ke Zhu* and Hanzhong Liu. Design-Based Theory for Causal Inference (in Chinese). *Advances in Mathematics (China)*. [Paper]

Submitted Papers and Working Papers

3. **Haoyang Yu**, Wei Ma and Hanzhong Liu*. Minimax Optimal Design with Spillover and Carryover Effects. [arXiv] [Code]
4. **Haoyang Yu**, Yilin Li, Lu Deng, Yong Wang, Xin Lu* and Hanzhong Liu*. Design-Based Edge-Level Causal Inference with Machine Learning-Assisted Covariate Adjustment. [arXiv]
5. **Haoyang Yu**, Anqi Zhao and Hanzhong Liu*. Unbiased Treatment Effect Estimation under Network Interference via Neighborhood-Excluded Cross-Fitting. [arXiv]
6. **Haoyang Yu**, Xin Lu* and Hanzhong Liu*. Calibrated Regression Adjustment in Completely Randomized Experiments with Moderately High-Dimensional Covariates.

Presentations

1. Minimax Optimal Design with Spillover and Carryover Effects. *The 9th National Academic Forum for Doctoral Students in Statistics*, Guangzhou, November 2025.
2. Minimax Optimal Design with Spillover and Carryover Effects. *The 3rd Joint Conference on Statistics and Data Science in China*, Hangzhou, July 2025.
3. Sharp Variance Estimator and Causal Bootstrap in Stratified Randomized Experiments. *Academic Symposium of the Beijing Biometrics Association*, Beijing, January 2025.
4. Minimax Optimal Design with Spillover and Carryover Effects. *Tsinghua Statistics Colloquium*, Beijing, China, November 2024.

Awards

Merit Student of Tsinghua University	2026
Outstanding Student Cadre of Tsinghua University	2019, 2025, 2026
Outstanding Teaching Assistant of Tsinghua University	2025
12·9 Counselor Scholarship of Tsinghua University	2025
First Prize at the National Academic Forum for Doctoral Students in Statistics	2025
Second Prize at the Academic Symposium of the Beijing Biometrics Association	2025
Comprehensive Scholarship of Tsinghua University	2019–2021, 2023, 2025, 2026
Outstanding Graduate of Tsinghua University	2022
Chinese National Scholarship	2021

Teaching

Teaching Assistant for Undergraduate Courses

1. Statistical Inference, Tsinghua University, Fall 2022
2. Introduction to Bayesian Statistics, Tsinghua University, Spring 2023
3. Introduction to Causal Inference, Tsinghua University, Fall 2023
4. Introduction to Bayesian Statistics, Tsinghua University, Spring 2024
5. Introduction to Causal Inference, Tsinghua University, Fall 2024
   - Awarded Outstanding Teaching Assistant of Tsinghua University (2025)
6. Introduction to Biostatistics and Bioinformatics, Tsinghua University, Spring 2026

Teaching Assistant for Graduate Courses

7. Advanced Probability Theory II, Tsinghua University, Spring 2024
8. Selected Topics on Advanced Statistics, Tsinghua University, Spring 2026
   - Topic: Bayesian Theory, Methods, and Computation

Software

`CausalBootstrap`: R package for sharp variance estimation and bootstrap inference in stratified randomized experiments.

Student Affairs

Student Assistant, Department of Statistics and Data Science, Tsinghua University 2024 – 2027

Graduate Student Affairs Assistant, Department of Statistics and Data Science, Tsinghua University 2024 – 2027

Undergraduate Student Counselor, Weiyang College, Tsinghua University 2022 – 2026

President, Student Union, Department of Physics, Tsinghua University 2020 – 2021

President, Student Association for Science and Technology, Weiyang College, Tsinghua University 2020 – 2022

Skills

R, Python, MATLAB, Mathematica, LaTeX.